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Tutorial Course for Intro to Sensor-Integrated Systems

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Reliability Assessment

- **Reliability $R(t)$:** The probability that a system operates normally and **achieves the specified performance** within a defined environment and time frame.
- **Unreliability $F(t)$:** The probability that the system **fails to achieve** the specified performance under the aforementioned conditions.
- **Golden Rule:** Equipment is either good or bad, therefore $R(t) + F(t) = 1$

To measure equipment quality, we need to focus on three core time periods:

- **MTTF (Mean Time To Failure):** For *non-repairable* components, this calculates the "average lifespan."
- **MDT (Mean Down Time):** For *repairable* components, this calculates "how long each repair takes on average."
 - $MDT = \text{Total Down Time} / \text{Number of Failures}$
- **MTBF (Mean Time Between Failures):** For *repairable* components, this calculates "how long it can be used between two failures."

$$MTBF = \text{Total Up Time} / \text{Number of Failures}$$

Based on the times above, we derive two ultimate metrics to measure the system:

- **Mean Failure Rate ($\bar{\lambda}$):** The frequency of damage per unit of time.
Formula: $\bar{\lambda} = 1/MTBF$
- **Availability (A):** The probability that the system "doesn't drop the ball" (fails) at critical moments (the proportion of uptime).

$$\text{Formula: } A = \frac{MTBF}{MTBF + MDT}$$

Exercise

A batch of 100 identical thermocouples were tested over a 12-week period. Twenty failures were recorded and the corresponding down times in hours were as follows:

5, 6, 7, 8, 4, 7, 8, 10, 5, 4, 8, 5, 4, 5, 6, 5, 4, 9, 8, 6

Calculate:

- (a) Mean down time
- (b) Mean time between failures
- (e) Mean failure rate
- (d) Availability.

Solution

Summary of Given Data

- **Total number of tested items (N):** 100
- **Total test duration (T):** 12 weeks. Converted to hours: $T = 12 \times 7 \times 24 = 2016$ hours
- **Total number of failures (N_F):** 20
- **Total down time ($\sum T_{Dj}$):** Summing up the 20 down times listed in the problem:
 $5 + 6 + 7 + 8 + 4 + 7 + 8 + 10 + 5 + 4 + 8 + 5 + 4 + 5 + 6 + 5 + 4 + 9 + 8 + 6 = 124$ hours

(a) Mean down time (MDT)

Mean down time is equal to the total down time divided by the number of failures.

$$\text{MDT} = \frac{\text{Total down time}}{N_F} = \frac{124}{20} = \mathbf{6.2 \text{ hours}}$$

(b) Mean time between failures (MTBF)

First, we need to calculate the **total up time** of these 100 thermocouples during the test period, which is the theoretical total test time minus the total down time, and then divide by the number of failures.

$$\text{Total test time} = N \times T = 100 \times 2016 = 201,600 \text{ hours}$$

$$\text{Total up time} = 201,600 - 124 = 201,476 \text{ hours}$$

$$\text{MTBF} = \frac{\text{Total up time}}{N_F} = \frac{201,476}{20} = \mathbf{10,073.8 \text{ hours}}$$

(c) Mean failure rate ($\bar{\lambda}$)

$$\bar{\lambda} = \frac{1}{\text{MTBF}}$$

$$\bar{\lambda} = \frac{1}{10073.8} \approx \mathbf{9.927 \times 10^{-5} \text{ failures/hour}}$$

(d) Availability (A)

$$A = \frac{\text{MTBF}}{\text{MTBF} + \text{MDT}}$$

$$A = \frac{10073.8}{10073.8 + 6.2} = \frac{10073.8}{10080} \approx \mathbf{0.99938 \text{ (or } 99.938\%)}$$

System Architecture Reliability Assessment

Constant Failure Rate Model

For components in their normal useful life period, the failure rate λ is constant.

- **Reliability Exponential Decay Formula:** $R(t) = e^{-\lambda t}$
- **Unreliability:** $F(t) = 1 - e^{-\lambda t}$

Series Systems

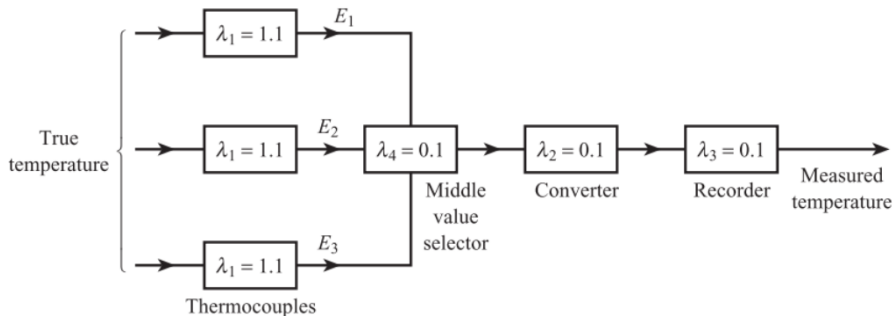
- **Logic:** All components must work normally at the same time for the system to operate.
- **Calculation Rules:**
 - **Multiplication of Reliability:** $R_{SYST} = R_1 \times R_2 \times \dots \times R_n$
 - **Addition of Failure Rates (Most Common):** $\lambda_{SYST} = \lambda_1 + \lambda_2 + \dots + \lambda_n$
- **Best Practice Guide:** The more components in series, the higher the total failure rate! Design should follow the principle: "Do not multiply entities beyond necessity."

Parallel Systems

- **Logic:** The system can operate as long as **at least one** component is working normally (it only fails if everything fails).
- **Calculation Rules:** Convert to calculating "**Unreliability F** ".
 - **Multiplication of Unreliability:** $F_{SYST} = F_1 \times F_2 \times \dots \times F_n$
 - **Shortcut for Identical Components:** $F_{SYST} = F^n$
- **Best Practice Guide:** Parallelism is a "heavy hitter" for improving reliability (probabilities get smaller as they are multiplied), but the trade-off is doubled costs.

Exercise

A more cost-effective **mixed redundancy system** is now designed: three thermocouples are used in parallel (output via a median selector), which are then connected in series with one converter and one recorder. Find the total reliability ($R_{OVERALL}$) and unreliability ($F_{OVERALL}$) of this mixed system at $t = 0.5$ years.



Solution

1. Calculate the equivalent unreliability and reliability of the parallel thermocouple section:

The unreliability of a single thermocouple ($\lambda_1 = 1.1$) at $t = 0.5$ is $1 - e^{-1.1 \times 0.5} = 1 - e^{-0.55}$.

The unreliability after connecting three in parallel is $F_1 = (1 - e^{-0.55})^3 = 0.076$.

The equivalent reliability is $R_1 = 1 - 0.076 = 0.924$.

2. Calculate the reliability of the remaining series components (converter, recorder, etc., assuming $\lambda = 0.1$):

$$R_2 = R_3 = R_4 = e^{-0.1 \times 0.5} = e^{-0.05} = 0.951.$$

3. Calculate the overall system reliability:

The entire system can be viewed as a series connection of these equivalent modules:

$$R_{OVERALL} = R_1 R_2 R_3 R_4 = 0.924 \times (0.951)^3 = 0.795$$

The total unreliability $F_{OVERALL} = 1 - 0.795 = 0.205$.

Exercise

A flow measurement system consists of an orifice plate ($\lambda = 0.75$), differential pressure transmitter ($\lambda = 1.0$), square root extractor ($\lambda = 0.1$) and recorder ($\lambda = 0.1$). Calculate the probability of losing the flow measurement after 0.5 year for the following:

- (a) A single flow measurement system
- (b) Three identical flow measurement systems in parallel
- (c) A system with three orifice plates, three differential pressure transmitters and a middle value selector relay ($\lambda = 0.1$). The selected transmitter output is passed to a single square root extractor and recorder.

Annual failure rate data are given in brackets; assume that all systems were initially checked and found to be working correctly.

Solution

Based on the information provided, we first list the annual failure rates (λ , unit: failures/year) and the operating time (t , unit: years) for each component:

- Orifice plate: $\lambda_1 = 0.75$
- Recorder: $\lambda_4 = 0.1$
- Differential pressure transmitter: $\lambda_2 = 1.0$
- Middle value selector relay: $\lambda_5 = 0.1$
- Square root extractor: $\lambda_3 = 0.1$
- Operating time: $t = 0.5$ years

In a constant failure rate model, a component's reliability is $R(t) = e^{-\lambda t}$, and the probability of failure (unreliability) is $F(t) = 1 - e^{-\lambda t}$.

(a) Single Flow Measurement System

This is a purely **series system**. If any single component fails, the entire system loses measurement data. The total failure rate of a series system equals the sum of the individual component failure rates:

$$\begin{aligned}\lambda_s &= \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 \\ \lambda_s &= 0.75 + 1.0 + 0.1 + 0.1 = 1.95 \text{ year}^{-1}\end{aligned}$$

System reliability R_a after 0.5 years of operation:

$$R_a = e^{-1.95 \times 0.5} = e^{-0.975} \approx 0.3772$$

Therefore, the **probability of losing measurement (unreliability F_a)** is:

$$F_a = 1 - R_a = 1 - 0.3772 = \mathbf{0.6228} \text{ (or } \mathbf{62.28\%})$$

Solution

(b) Three Identical Flow Measurement Systems in Parallel

This is a **parallel redundancy system**. The system only loses measurement data when all three single systems fail simultaneously. According to the formula for parallel systems, the total unreliability equals the product of the unreliability of each branch. We already calculated the unreliability of a single system in part (a): $F_a = 0.6228$.

Unreliability of three systems in parallel (F_b):

$$F_b = (F_a)^3 = (0.6228)^3 \approx \mathbf{0.2416} \text{ (or } \mathbf{24.16\%})$$

(c) Mixed Redundancy System

This system is more complex and can be broken down into two parts for analysis:

Part 1: Front-end Parallel Subsystem

This consists of 3 parallel branches, each made of an orifice plate and a differential pressure transmitter in series. Per the textbook's simplified method for "middle value selector" front-end parallel circuits (see Section 7.1.5), this part is treated as a single redundant module:

- Failure rate of a single branch: $\lambda_{\text{branch}} = \lambda_1 + \lambda_2 = 0.75 + 1.0 = 1.75 \text{ year}^{-1}$
- Unreliability of a single branch: $F_{\text{branch}} = 1 - e^{-1.75 \times 0.5} = 1 - e^{-0.875} \approx 1 - 0.4169 = 0.5831$
- Unreliability of the entire front-end: $F_{\text{front}} = (F_{\text{branch}})^3 = (0.5831)^3 \approx 0.1983$
- Reliability of the front-end module: $R_{\text{front}} = 1 - 0.1983 = 0.8017$

Solution

Part 2: Back-end Series Subsystem

The signal from the selector passes through the middle value selector relay, square root extractor, and recorder. These three components are in series.

- Total failure rate of the back-end series: $\lambda_{\text{back}} = \lambda_5 + \lambda_3 + \lambda_4 = 0.1 + 0.1 + 0.1 = 0.3 \text{ year}^{-1}$
- Reliability of the back-end: $R_{\text{back}} = e^{-0.3 \times 0.5} = e^{-0.15} \approx 0.8607$

Calculating the Total System:

The entire system is equivalent to "Part 1" and "Part 2" in series.

$$R_c = R_{\text{front}} \times R_{\text{back}}$$

$$R_c = 0.8017 \times 0.8607 \approx 0.6900$$

Probability of losing measurement (unreliability F_c):

$$F_c = 1 - R_c = 1 - 0.6900 = \mathbf{0.3100} \text{ (or } \mathbf{31.00\%})$$

The TLOC Assessment Method

Total Life Cycle Operating Cost (TLOC)

For any given application, the best system is the one with the **minimum TLOC**.

$$TLOC = InitialPurchaseCost + TotalFailureMaintenanceCost + TotalMeasurementErrorCost$$

Process Downtime Loss (C_p)

- **Formula Analysis:** Cost of a Single Failure = Spare Parts/Labor + $C_p \times RepairTime$
- The loss caused by a factory shutdown due to equipment failure is often hundreds or thousands of times the cost of a spare part! "Repairing quickly" and "not breaking easily" are extremely valuable.

Measurement Error Penalty Cost ($C(E)$)

- The equipment may not be broken, but it may be inaccurate (e.g., mixing ratios out of balance in a chemical plant, or low flow in an oil pipeline). The hidden losses saved by high-precision equipment often lead to a rapid return on investment.

Exercise

Use the data given in Table Prob. 3 to decide which level measurement system should be purchased. Assume a breakdown maintenance only strategy is practised, each system has the same measurement error, and there is a 10-year total lifetime.

	Parameter	System 1	System 2
Initial cost	£	1000	2000
Materials cost per repair	£	20	15
Labour cost per hour	£	10	10
Process cost per hour	£	100	100
Repair time	h	8	12
Annual failure rate	yr ⁻¹	2.0	1.0

Solution

Based on the conditions provided in the problem:

- 1. Breakdown maintenance only:** This means there are no periodic preventive maintenance costs.
- 2. Identical measurement errors:** This means the potential financial losses caused by measurement errors for both systems are the same and can be ignored during the comparison.
- 3. Total Lifespan (T):** 10 years.

Therefore, the TLOC formula for comparing the two systems can be simplified to:

$$TLOC = InitialCost + TotalFailureMaintenanceCostoverLifespan$$

Where:

- Total failures over lifespan = Annual failure rate (λ) $\times$ Total lifespan (T)
- Cost per failure = Material cost (C_R) + [Labor cost (C_L) + Process cost (C_P)] $\times$ Repair time (T_R)

Thus, the simplified formula is:

$$TLOC = C_1 + [C_R + (C_L + C_P) \times T_R] \times \lambda \times T$$

System 1

- **Initial Cost (C_1):** £1000
- **Cost per failure:**
 $£20 + (£10 + £100) \times 8 h = 20 + 110 \times 8 = £900$
- **Total failures over 10 years:**
 $2.0 failures/year \times 10 years = 20 failures$
- **Total maintenance cost over lifespan:**
 $£900 \times 20 = £18,000$
- **TLOC of System 1:**
 $TLOC_1 = 1000 + 18000 = £19,000$

System 2

- **Initial Cost (C_1):** £2000
- **Cost per failure:**
 $£15 + (£10 + £100) \times 12 h = 15 + 110 \times 12 = £1,335$
- **Total failures over 10 years:**
 $1.0 failures/year \times 10 years = 10 failures$
- **Total maintenance cost over lifespan:**
 $£1,335 \times 10 = £13,350$
- **TLOC of System 2:**
 $TLOC_2 = 2000 + 13350 = £15,350$

Capacitive Sensing Components

1. Core Physical Model and Formula

➤ **Fundamental Formula:** $C = \frac{\epsilon_0 \epsilon_r A}{d}$

Capacitance C depends on: vacuum permittivity ϵ_0 , relative permittivity ϵ_r , plate overlap area A , and plate separation distance d

➤ **Three Main Types:** Variable distance type (change d), variable area type (change A), and **variable dielectric constant type** (change ϵ_r).

2. Key Focus: Variable Dielectric Constant Sensor

➤ **Equivalent Circuit Principle:** When a dielectric plate is inserted or pulled from between the plates, the sensor is equivalent to **two capacitors in parallel** (Air section + Dielectric section).

➤ **Total Capacitance Formula:**

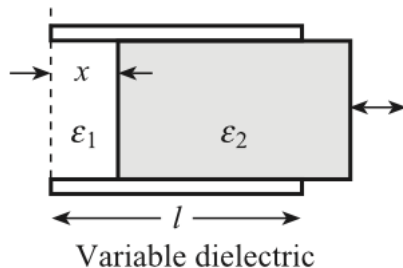
$$C_{total} = C_{air} + C_{dielectric}$$

$$C_{total} = \frac{\epsilon_0 W}{d} [\epsilon_1 x + \epsilon_2 (l - x)]$$

➤ **Measurement Characteristic:** There is a **perfectly linear relationship** between capacitance C and displacement x .

Exercise

A variable dielectric capacitive displacement sensor consists of two square metal plates of side 5 cm, separated by a gap of 1 mm. A sheet of dielectric material 1 mm thick and of the same area as the plates can be slid between them as shown in Figure 8.9. Given that the dielectric constant of air is 1 and that of the dielectric material 4, calculate the capacitance of the sensor when the input displacement $x = 0.0, 2.5$ and 5.0 cm.



Solution

1. Extract Known Parameters

- **Plate Dimensions:** Square, side length $w = l = 5 \text{ cm} = 0.05 \text{ m}$
- **Plate Separation:** $d = 1 \text{ mm} = 0.001 \text{ m}$
- **Dielectric Thickness:** 1 mm (equal to the plate separation, meaning the dielectric completely fills the gap vertically)
- **Relative Permittivity of Air:** $\varepsilon_1 = 1$
- **Relative Permittivity of Dielectric:** $\varepsilon_2 = 4$
- **Vacuum Permittivity:** $\varepsilon_0 = 8.85 \text{ pF/m} = 8.85 \times 10^{-12} \text{ F/m}$
- **Displacements to be calculated (x):** 0.0 cm , 2.5 cm , and 5.0 cm respectively.

2. Formula Derivation

When the dielectric plate is pulled out by a distance x :

- **Area covered by the air section:** $A_1 = w \cdot x$
- **Remaining area covered by the dielectric:** $A_2 = w \cdot (l - x)$

The total capacitance C is the sum of these two parallel capacitors (per textbook formula [8.20]):

Solution

$$C = C_{\text{air}} + C_{\text{dielectric}} = \frac{\varepsilon_0 \varepsilon_1 A_1}{d} + \frac{\varepsilon_0 \varepsilon_2 A_2}{d}$$

$$C = \frac{\varepsilon_0 w}{d} [\varepsilon_1 x + \varepsilon_2 (l - x)]$$

Substituting the relative permittivities $\varepsilon_1 = 1$ and $\varepsilon_2 = 4$:

$$C = \frac{\varepsilon_0 w}{d} [x + 4(l - x)] = \frac{\varepsilon_0 w}{d} [4l - 3x]$$

First, let's calculate the constant coefficient at the front of the formula:

$$\frac{\varepsilon_0 w}{d} = \frac{8.85 \times 10^{-12} \text{ F/m} \times 0.05 \text{ m}}{0.001 \text{ m}} = 4.425 \times 10^{-10} \text{ F/m} = 442.5 \text{ pF/m}$$

3. Calculate Capacitance for Three Displacement Scenarios

Case (1): Displacement $x = 0.0 \text{ cm} = 0 \text{ m}$

$$C = 442.5 \times [4 \times 0.05 - 3 \times 0] = 442.5 \times 0.2$$

$$C = \mathbf{88.5 \text{ pF}}$$

Case (2): Displacement $x = 2.5 \text{ cm} = 0.025 \text{ m}$

$$C = 442.5 \times [4 \times 0.05 - 3 \times 0.025] = 442.5 \times [0.2 - 0.075] = 442.5 \times 0.125$$

$$C = \mathbf{55.3125 \text{ pF}}$$

Case (3): Displacement $x = 5.0 \text{ cm} = 0.05 \text{ m}$

$$C = 442.5 \times [4 \times 0.05 - 3 \times 0.05] = 442.5 \times [0.2 - 0.15] = 442.5 \times 0.05$$

$$C = \mathbf{22.125 \text{ pF}}$$

Variable Reluctance Sensing Components

1. Core Physical Model and Formulas

- **Ohm's Law for Magnetic Circuits:** $m.m.f. = \phi \times R$ (*Magnetomotive force = Magnetic flux $\times$ Reluctance*)
- **Relationship between Inductance and Reluctance:** $L = \frac{n^2}{R_{total}}$ (*Inductance L is inversely proportional to total reluctance*)
- **Reluctance Calculation Formula:** $R = \frac{l}{\mu\mu_0 A}$ (*Reluctance depends on path length l , area A , and material magnetic permeability μ*)

2. Key Focus: Variable Reluctance Displacement Sensor

- **Series Composition of Total Reluctance:**

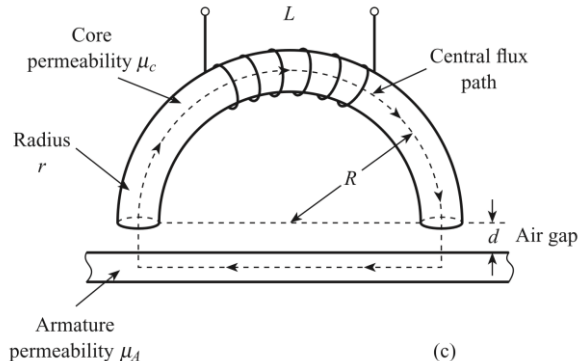
$$R_{total} = R_{core}(IronCore) + R_{armature}(Armature) + R_{gap}(AirGap)$$

- **Core Characteristics (Highly Non-linear):**

- For ferromagnetic materials, $\mu \approx 100 \sim 1000$, while for air, $\mu \approx 1$.
- **Conclusion:** Even if the air gap (d) is only 1 mm, its reluctance may be far greater than that of a thick iron core!
- A tiny increase in displacement d will cause the total reluctance to rise sharply, leading to a **non-linear plunge** in inductance L .

Exercise

A variable reluctance sensor consists of a core, a variable air gap and an armature. The core is a steel rod of diameter 1 cm and relative permeability 100, bent to form a semi-circle of diameter 4 cm. A coil of 500 turns is wound onto the core. The armature is a steel plate of thickness 0.5 cm and relative permeability 100. Assuming the relative permeability of air = 1.0 and the permeability of free space = $4\pi \times 10^{-7} \text{ H m}^{-1}$, calculate the inductance of the sensor for air gaps of 1 mm and 3 mm.



Solution

1. Extract Known Parameters and Symbol Mapping

- **Core:**
 - Cross-sectional diameter = 1 cm $\rightarrow$ radius $r = 0.5 \text{ cm} = 0.005 \text{ m}$
 - Semicircle radius = 4 cm $\rightarrow$ core mean radius $R = 2 \text{ cm} = 0.02 \text{ m}$
 - Relative magnetic permeability of the core $\mu_r = 100$
- **Coil:** Number of turns $n = 500$
- **Armature:**
 - Thickness $t = 0.5 \text{ cm} = 0.005 \text{ m}$
 - Relative magnetic permeability of the armature $\mu_A = 100$
- **Environment Constants:**
 - Relative magnetic permeability of air $\mu_{air} = 1.0$
 - Vacuum permeability $\mu_0 = 4\pi \times 10^{-7} \text{ H} \cdot \text{m}^{-1}$
- **Target Air Gaps (d):** $d_1 = 1 \text{ mm} = 0.001 \text{ m}$, $d_2 = 3 \text{ mm} = 0.003 \text{ m}$

Solution

2. Formula Derivation

According to the principles of magnetic circuits, the total reluctance $\mathcal{R}_{\text{TOTAL}}$ is composed of three parts in series:

$$\mathcal{R}_{\text{TOTAL}} = \mathcal{R}_{\text{CORE}} + \mathcal{R}_{\text{ARMATURE}} + \mathcal{R}_{\text{GAP}}$$

Substituting the reluctance calculation formulas provided in the textbook:

1. **Core Reluctance:** $\mathcal{R}_{\text{CORE}} = \frac{\pi R}{\mu_0 \mu_r (\pi r^2)} = \frac{R}{\mu_0 \mu_r r^2}$
2. **Armature Reluctance** (path length is $2r$): $\mathcal{R}_{\text{ARMATURE}} = \frac{2r}{\mu_0 \mu_A (2rt)} = \frac{r}{\mu_0 \mu_A rt}$
3. **Air Gap Reluctance** (two gaps, total length $2d$): $\mathcal{R}_{\text{GAP}} = \frac{2d}{\mu_0 (\pi r^2)}$

In this problem, $r = t = 0.005 \text{ m}$ and $\mu_r = \mu_A = 100$. Therefore, the core and armature reluctances are equal. We can define the fixed reluctance (excluding the air gap) as $\mathcal{R}_0$:

$$\mathcal{R}_0 = \mathcal{R}_{\text{CORE}} + \mathcal{R}_{\text{ARMATURE}} = 2 \times \frac{0.02}{(4\pi \times 10^{-7}) \times 100 \times (0.005)^2}$$

$$\mathcal{R}_0 = \frac{0.04}{100 \times 25 \times 10^{-6} \times 4\pi \times 10^{-7}} = \frac{4 \times 10^7}{\pi} \approx 12.732 \times 10^6 \text{ H}^{-1}$$

Solution

3. Calculating Inductance for Different Air Gaps

The final calculation formula for inductance is: $L = \frac{n^2}{\mathcal{R}_{\text{TOTAL}}}$

Scenario (1): When air gap $d = 1 \text{ mm} = 0.001 \text{ m}$

- **Calculate Gap Reluctance:**

$$\mathcal{R}_{\text{GAP}} = \frac{2 \times 0.001}{(4\pi \times 10^{-7}) \times \pi \times (0.005)^2} = \frac{0.002}{10^{-10}\pi^2} = \frac{2 \times 10^7}{\pi^2} \approx 20.264 \times 10^6 \text{ H}^{-1}$$

- **Calculate Total Reluctance:**

$$\mathcal{R}_{\text{TOTAL}} = (12.732 + 20.264) \times 10^6 = 32.996 \times 10^6 \text{ H}^{-1}$$

- **Calculate Inductance:**

$$L = \frac{500^2}{32.996 \times 10^6} \approx 0.00758 \text{ H} = \mathbf{7.58 \text{ mH}}$$

Scenario (2): When air gap $d = 3 \text{ mm} = 0.003 \text{ m}$

- **Calculate Gap Reluctance:**

Since the gap length has tripled, the gap reluctance also triples:

$$\mathcal{R}_{\text{GAP}} = 3 \times 20.264 \times 10^6 = 60.793 \times 10^6 \text{ H}^{-1}$$

- **Calculate Total Reluctance:**

$$\mathcal{R}_{\text{TOTAL}} = (12.732 + 60.793) \times 10^6 = 73.525 \times 10^6 \text{ H}^{-1}$$

- **Calculate Inductance:**

$$L = \frac{500^2}{73.525 \times 10^6} \approx 0.00340 \text{ H} = \mathbf{3.40 \text{ mH}}$$

Active Electrical Sensing Elements

1. Core Concept: What does "Active" mean?

- **Self-generating Mechanism:** Does not require an external power supply; it extracts energy directly from the measured object or system (such as mechanical kinetic energy).
- **Energy Conversion:** Directly converts physical motion into an AC electrical signal output.

2. Operating Principle: Faraday's Law of Induction

- **Fundamental Formula:** $E = -\frac{dN}{dt}$

(The induced electromotive force is equal to the negative rate of change of the magnetic flux linkage)

- **Physical Process:** Rotation of a ferromagnetic gear → Periodic change in the air-gap magnetic flux between the teeth and magnetic poles → Fluctuations in the magnetic flux linkage N within the coil → Generation of AC voltage E .

3. Mathematical Model and Core Conclusions

- **Magnetic Flux Equation:** The flux linkage varies approximately sinusoidally with the angular displacement θ :

$$N(\theta) \approx a + b \cos(m\theta) \quad (\text{where } m \text{ is the number of gear teeth})$$

- **Output Signal Equation:** Derived using the chain rule of calculus to find the AC electrical signal:

$$E = (bm\omega_r) \sin(m\omega_r t) \quad (\text{where } \omega_r \text{ is the angular velocity of the gear, i.e., } d\theta/dt)$$

The Two "Direct Proportionality" Laws:

- **Amplitude Proportionality:** The peak value of the output voltage, $\hat{E} = bm\omega_r$, is directly proportional to the rotational speed.
- **Frequency Proportionality:** The frequency of the AC signal, $f = \frac{m\omega_r}{2\pi}$, is directly proportional to the rotational speed.
(Engineering shortcut formula: $f = \frac{RPM}{60} \times m$ (number of teeth))

Exercise

A variable reluctance tachogenerator consists of a ferromagnetic gear wheel with 22 teeth rotating close to a magnet and coil assembly. The total flux N linked by the coil is given by:

$$N(\theta) = 4.0 + 1.5 \cos 22\theta \text{ milliwebers}$$

where θ is the angular position of the wheel relative to the axis of the magnet. Calculate the amplitude and frequency of the output signal when the angular velocity of the wheel is 1000 and 10 000 r.p.m.

Solution

1. Extract Known Parameters

- **Number of teeth (m):** 22
- **Equation for flux linkage ($N(\theta)$):** $N(\theta) = 4.0 + 1.5 \cos(22\theta)$ mWb
(Note the unit conversion: $1 \text{ mWb} = 10^{-3} \text{ Wb}$, so the varying flux amplitude $b = 1.5 \times 10^{-3} \text{ Wb}$)
- **Given rotational speeds:** 1000 r.p.m. and 10000 r.p.m. (revolutions per minute)

2. Theoretical Formula Derivation

According to Faraday's Law, the induced electromotive force E (output signal) in the coil is the negative rate of change of flux linkage over time:

$$E = -\frac{dN}{dt}$$

Using the **Chain Rule** from calculus, we expand this as the product of the derivative with respect to angular displacement θ and the angular velocity ω_r :

$$E = -\frac{dN}{d\theta} \cdot \frac{d\theta}{dt}$$

$$E = -[1.5 \times 10^{-3} \times (-\sin 22\theta) \times 22] \cdot \omega_r$$

Given angular velocity $\frac{d\theta}{dt} = \omega_r$ (rad/s), we substitute the flux equation:

$$E = 0.033\omega_r \sin(22\theta)$$

Since angular displacement $\theta = \omega_r t$, the final expression for the AC output signal is:

$$E = -\left[\frac{d}{d\theta}(4.0 + 1.5 \cos 22\theta) \times 10^{-3}\right] \cdot \omega_r$$

$$E = 0.033\omega_r \sin(22\omega_r t)$$

Solution

From this, we extract the two key parameter calculation formulas:

- **Amplitude ($\hat{E}$):** $\hat{E} = 0.033\omega_r$ (V)
- **Frequency (f):** The angular frequency of the signal is $22\omega_r$, so the actual frequency is $f = \frac{22\omega_r}{2\pi}$ (Hz).
(A faster engineering calculation: $f = \frac{\text{revolutions per second}}{1} \times \text{teeth, i.e., } f = \frac{\text{RPM}}{60} \times m$)

3. Numerical Calculation

Case (1): When speed is 1000 r.p.m.

First, convert the speed to standard angular velocity ω_r (rad/s): $\omega_r = 1000 \times \frac{2\pi}{60} = \frac{100\pi}{3} \approx 104.72$ rad/s

- **Calculate Amplitude $\hat{E}$:** $\hat{E} = 0.033 \times \frac{100\pi}{3} = 1.1\pi \approx \mathbf{3.456}$ V
- **Calculate Frequency f :** $f = \frac{1000}{60} \times 22 = \frac{2200}{60} \approx \mathbf{366.67}$ Hz

Case (2): When speed is 10000 r.p.m.

Since the speed is exactly 10 times that of Case (1), and based on our derivation, both amplitude and frequency have a strict **direct proportionality** with speed, the results are simply scaled by 10:

- $\omega_r = 10000 \times \frac{2\pi}{60} = \frac{1000\pi}{3} \approx 1047.2$ rad/s
- **Amplitude $\hat{E}$:** $\hat{E} = 1.1\pi \times 10 = 11\pi \approx \mathbf{34.56}$ V
- **Frequency f :** $f = 366.67 \times 10 \approx \mathbf{3666.67}$ Hz

Mechanical Mechanics & Elastic Deformation

1. Stress and Strain

➤ Stress (Stress, σ) — The "Cause" of deformation:

- **Definition:** Internal force sustained per unit area.
- **Formula:** $\sigma = F/A$ (Tensile force is positive +, compressive force is negative -).

➤ Strain (Strain, e) — The "Result" of deformation:

- **Definition:** The relative change in the dimensions of an object (dimensionless).
- **Formula:** $e = \Delta l/l$.

➤ Hooke's Law:

- **Relationship:** Young's Modulus $E = \sigma/e$ (Describes a material's ability to resist elastic deformation/stiffness).

2. 3D Deformation Linkage Rule: Poisson's Effect

➤ Physical Phenomenon: When a material is stretched (lengthened), it must become thinner; when compressed (shortened), it must become thicker.

➤ Core Formula:

$$e_T = -\nu e_L$$

(Transverse strain $e_T = -(\text{Poisson's ratio}) \nu \times (\text{longitudinal strain}) e_L$)

- The **negative sign (-)** in the formula is vital! It indicates that the directions of transverse and longitudinal deformation are **always opposite**.
- If subjected to **compressive force** → longitudinal strain e_L is negative → a negative times a negative equals a positive, so transverse strain e_T is positive (the material bulges and thickens).

Exercise

A steel bar has a cross-sectional area of 10^{-3} m^2 , Young's modulus of $2 \times 10^{11} \text{ Pa}$ and Poisson's ratio of 0.4. If the bar is subject to a compressive force of 10^5 N , find the corresponding longitudinal and transverse strains.

Solution

1. Extract Known Parameters and Symbol Definitions

- Cross-sectional area (A): 10^{-3} m^2
- Young's modulus (E): $2 \times 10^{11} \text{ Pa}$ (or N/m^2)
- Poisson's ratio (ν): 0.4
- Applied Force (F): -10^5 N

Step 1: Calculate Longitudinal Strain (e_L)

According to Hooke's Law, stress $\sigma = F/A$ and $\sigma = E \cdot e_L$. Therefore, the formula for longitudinal strain is:

$$e_L = \frac{\sigma}{E} = \frac{F}{AE}$$

Substituting the known values:

$$e_L = \frac{-10^5}{10^{-3} \times 2 \times 10^{11}}$$

$$e_L = \frac{-10^5}{2 \times 10^8}$$

$$e_L = -0.5 \times 10^{-3}$$

$$e_L = -5 \times 10^{-4}$$

Step 2: Calculate Transverse Strain (e_T)

According to the definition of Poisson's ratio, as a material shortens longitudinally under compression, it simultaneously expands transversally (bulges). The relationship is given by formula [8.9] in the textbook:

$$e_T = -\nu e_L$$

Substituting the known values:

$$e_T = -0.4 \times (-5 \times 10^{-4})$$

$$e_T = 2 \times 10^{-4}$$

Dynamic Characteristics of Sensors

1. Physical Limitations of Dynamic Measurement

- **Ideal Sensor:** The output signal instantaneously and perfectly follows the changes in the input physical quantity.
- **Real-world Sensor:** All elastic sensing elements (such as accelerometers and pressure diaphragms) physically consist of three basic elements, forming a **second-order mechanical system**:
 - **Mass / Moment of Inertia (m / I):** Possesses inertia and resists sudden changes in the state of motion.
 - **Damping / Friction (λ / b):** Dissipates energy and hinders relative motion.
 - **Stiffness / Elasticity (k / c):** Provides restorative force or torque during deformation.

2. Three Core Parameters Describing Dynamic Response

The dynamic transfer function $G(s)$ of the system is entirely determined by the following three parameters:

- **Steady-state Gain (K):** The measurement sensitivity under static conditions.
- **Natural Frequency (ω_n):** The "vibrational DNA" of the system itself (Translational $\omega_n = \sqrt{k/m}$, Rotational $\omega_n = \sqrt{c/I}$).
- **Damping Ratio (ξ):** Determines how quickly or slowly oscillations decay after the system is disturbed.

Response and Accelerometer Analysis

1. The Physical Essence of Accelerometer Measurement: Relative Displacement

- The outer casing of the accelerometer vibrates along with the object being measured.
- The internal inertial mass block lags behind due to inertia, resulting in a **relative displacement** (θ or x) between the mass block and the casing. The sensor measures exactly this relative displacement.

2. Amplitude-Frequency Characteristic Formula

At a sinusoidal input of angular frequency ω , the relationship between the relative displacement amplitude $\hat{\theta}$ and the base input amplitude $\widehat{\phi}_i$ is given by:

$$\frac{\hat{\theta}}{\widehat{\phi}_i} = \frac{(\omega/\omega_n)^2}{\sqrt{[1 - (\omega/\omega_n)^2]^2 + (2\xi\omega/\omega_n)^2}}$$

(Note: ω/ω_n is called the frequency ratio).

3. Engineering's "Golden Damping Ratio": $\xi = 0.707 (1/\sqrt{2})$

- **Physical Significance:** This ratio allows the sensor to maintain a flat response over the widest possible frequency range, neither producing sharp resonance peaks nor responding too sluggishly.
- **Mathematical Trick (Problem-Solving Secret):** When $\xi = 1/\sqrt{2}$, the complex denominator of the formula above can be perfectly simplified to:

$$\sqrt{1 + (\omega/\omega_n)^4}$$

This significantly reduces calculation difficulty!

Exercise

An accelerometer is to measure the angular acceleration of a rotating mixing vessel. The angular position of the vessel varies sinusoidally with time with amplitude 2.5 rad and period 2 s. The rotating seismic mass is equivalent to a mass of 0.1 kg on a weightless arm of length 5 cm. The stiffness of the spring is $2.5 \times 10^{-2} \text{ N m rad}^{-1}$ and the damping ratio is $1/\sqrt{2}$. The angular position of the seismic mass is measured with a secondary potentiometric sensor. What should the input range of this sensor be?

Solution

1. Extracted Parameters and Physical Model

- **Input Vibration (Base) Parameters:**

- Amplitude of angular displacement: $\hat{\phi}_i = 2.5 \text{ rad}$
- Vibration period: $T = 2 \text{ s} \rightarrow \text{Source angular frequency } \omega = \frac{2\pi}{T} = \pi \text{ rad/s}$

- **Internal Sensor (Angular Accelerometer) Parameters:**

- Equivalent inertial mass: $m = 0.1 \text{ kg}$
- Arm length: $r = 5 \text{ cm} = 0.05 \text{ m}$
- Spring torsional stiffness: $c = 2.5 \times 10^{-2} \text{ N} \cdot \text{m} \cdot \text{rad}^{-1}$
- Damping ratio: $\xi = 1/\sqrt{2}$ (This is the engineering **Golden Damping Ratio**)

2. Core System Parameter Calculation

Step 1: Calculate the Moment of Inertia (I)

$$I = mr^2 = 0.1 \times (0.05)^2 = 2.5 \times 10^{-4} \text{ kg} \cdot \text{m}^2$$

Step 2: Calculate the Natural Angular Frequency (ω_n)

Based on the second-order dynamic formula:

$$\omega_n = \sqrt{\frac{c}{I}} = \sqrt{\frac{2.5 \times 10^{-2}}{2.5 \times 10^{-4}}} = \sqrt{100} = 10 \text{ rad/s}$$

Step 3: Calculate the Frequency Ratio (x)

$$x = \frac{\omega}{\omega_n} = \frac{\pi}{10} \approx 0.31416$$

Solution

3. Transfer Function and "Golden Simplification" (The Technical Highlight)

According to the accelerometer equation [8.19] in the textbook: $I\ddot{\theta} + b\dot{\theta} + c\theta = -I\ddot{\phi}_i$. This converts to the amplitude ratio formula for frequency response:

$$\frac{\hat{\theta}}{\hat{\phi}_i} = \frac{(\omega/\omega_n)^2}{\sqrt{[1 - (\omega/\omega_n)^2]^2 + (2\xi\omega/\omega_n)^2}} = \frac{x^2}{\sqrt{(1 - x^2)^2 + (2\xi x)^2}}$$

Since the problem specifies $\xi = 1/\sqrt{2}$, the cross-term in the denominator simplifies as follows:

$$(2\xi x)^2 = (2 \times \frac{1}{\sqrt{2}} \times x)^2 = (\sqrt{2}x)^2 = 2x^2$$

Now, look inside the square root:

$$(1 - x^2)^2 + 2x^2 = (1 - 2x^2 + x^4) + 2x^2 = 1 + x^4$$

The complex amplitude ratio formula is simplified to:

$$\frac{\hat{\theta}}{\hat{\phi}_i} = \frac{x^2}{\sqrt{1 + x^4}}$$

4. Final Numerical Substitution

Calculate the values based on the simplified formula:

- $x^2 = (\pi/10)^2 \approx 0.0987$
- $x^4 = (\pi/10)^4 \approx 0.0097$
- Denominator = $\sqrt{1 + 0.0097} = \sqrt{1.0097} \approx 1.0048$

Calculate the final relative displacement amplitude $\hat{\theta}$:

$$\hat{\theta} = \hat{\phi}_i \times \frac{x^2}{\sqrt{1 + x^4}} = 2.5 \times \frac{0.0987}{1.0048} \approx \mathbf{0.2456 \text{ rad}}$$